

CDS6334

VISUAL INFORMATION

PROCESSING

Course Information
Trimester 2610





Course Information

- › CDS6334 Visual Information Processing
- › Specialization Elective for Data Science
 - Elective for all others
- › Pre-requisites
 - Object-Oriented Programming and Data Structures
 - “Other pre-requisites”
 - › Decent programming skills
 - › Like problem-solving



Course Instructor

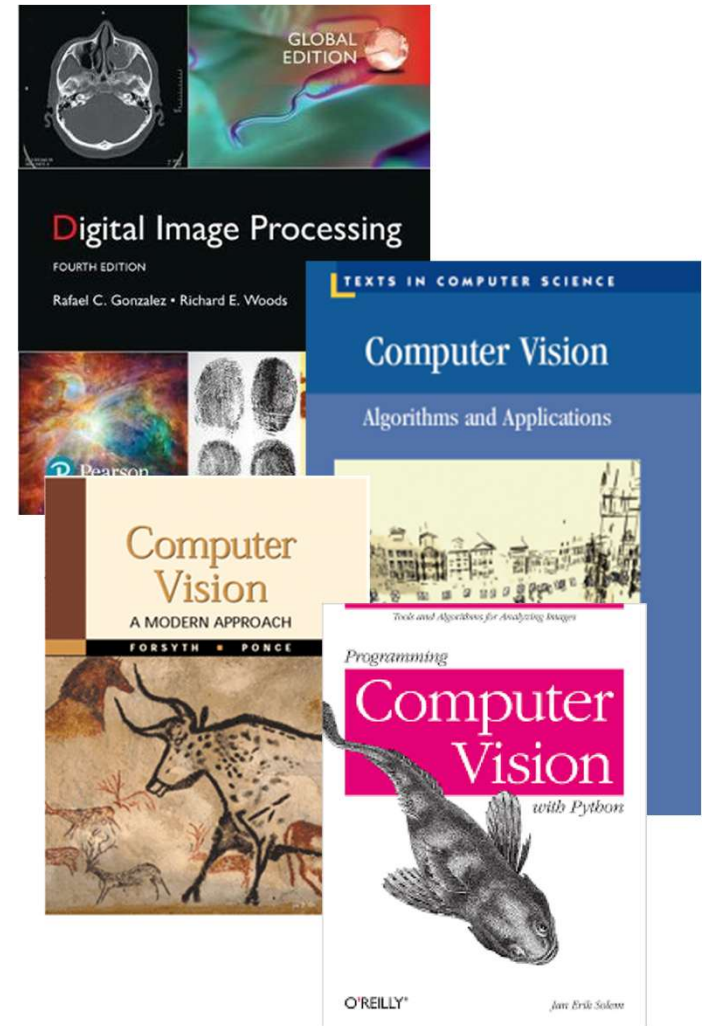
Dr. Syed Qamrun Nisa

- Email: qamrunnisa@mmu.edu.my
- Office: BR 2016



References

- › **Digital Image Processing**, 4th Ed. – Gonzalez & Woods (2018)
- › **Computer Vision: Algorithms and Applications** – R. Szeliski (2011)
- › **Computer Vision – A Modern Approach**, 2nd Ed., Forsyth & Ponce (2011)
- › **Programming Computer Vision with Python** – Solem (2012)
- › **Concise Computer Vision** – Klette (2014)





Topics Overview

1. Introduction to Images
2. Pixel Processing and Image Filtering
3. Low-level Features: Edges, Colour, Texture
4. Segmentation and Grouping
5. Feature Detectors and Descriptors
6. Deep Learning for Visual Information Processing
7. Video Processing: Motion
8. Visual Processing Applications



Software / Tools

- › Python 3 (3.14 latest)
 - Anaconda package [local installation]
(<https://www.anaconda.com/download>)
 - › Python + useful libraries like numpy, matplotlib, scipy, etc.)
 - › Jupyter Notebooks
 - › Spyder IDE
- › OpenCV – any 4.x version (4.12.0 latest)
 - Pip install opencv-python



Coursework Grading (100% Coursework)

› Quiz (20%)

- Problem sets/MCQ
- Additional quizzes, lab submission, etc. (TBA if applicable)

› Assignment (20%)

- Task specific individual assignment
- Programming + Report

› Test (20%) (5th June 2026, 6 pm – 8 pm)

- MCQ / discussion-based questions
- No programming

› Project (40%)

- Visual information system development / research
- Programming + Report + Presentation
- Group work



In order to pass this course, you need to score 40 % marks in both coursework component (quizzes, assignment, test) and final component (project).

Coursework component pass mark = 24/60

Final component pass marks = 16/40



Assessment Timeline

Trimester 2610 (Weeks)													
1	2	3	4	5	6	7	8	9	10	11	12	13	14
				Quiz 1					Test			Quiz 2	
				Assignment									
							Project						

Note: Assessment deadlines may subject to change depending on trimester arrangements.



Course Policies

› Late Policies

- **-10%** (of the submission mark) per day late
- Late due to **emergency cases** must be supported by **valid documentations** as proof (accident reports, hospitalization letters, etc.)
- **Disagreements** with groupmates should be **resolved before** submission (not used as an excuse to appeal for late submission)

› University-wide Policies

- **Plagiarism** is a serious offence. If found, immediate **ZERO** will be given to the assessment with no negotiation.
- **Cheating** during quizzes and tests will result in the submission being voided, and a **ZERO** will be given for the assessment.
- Attendance to be taken as per university rules (**barred if below 80%**)



Recommended Python Primers

- › Justin Johnson (Stanford Uni)'s Python Numpy tutorial
 - <http://cs231n.github.io/python-numpy-tutorial/>
 - iPython notebook:
 - › <https://github.com/kuleshov/cs228-material/blob/master/tutorials/python/cs228-python-tutorial.ipynb>
- › SciPy Lecture Notes
 - One document to learn numerics, science, and data with Python
 - <https://lectures.scientific-python.org/>
- › OpenCV Python Tutorials (make sure version is correct)
 - https://docs.opencv.org/4.9.0/d6/d00/tutorial_py_root.html

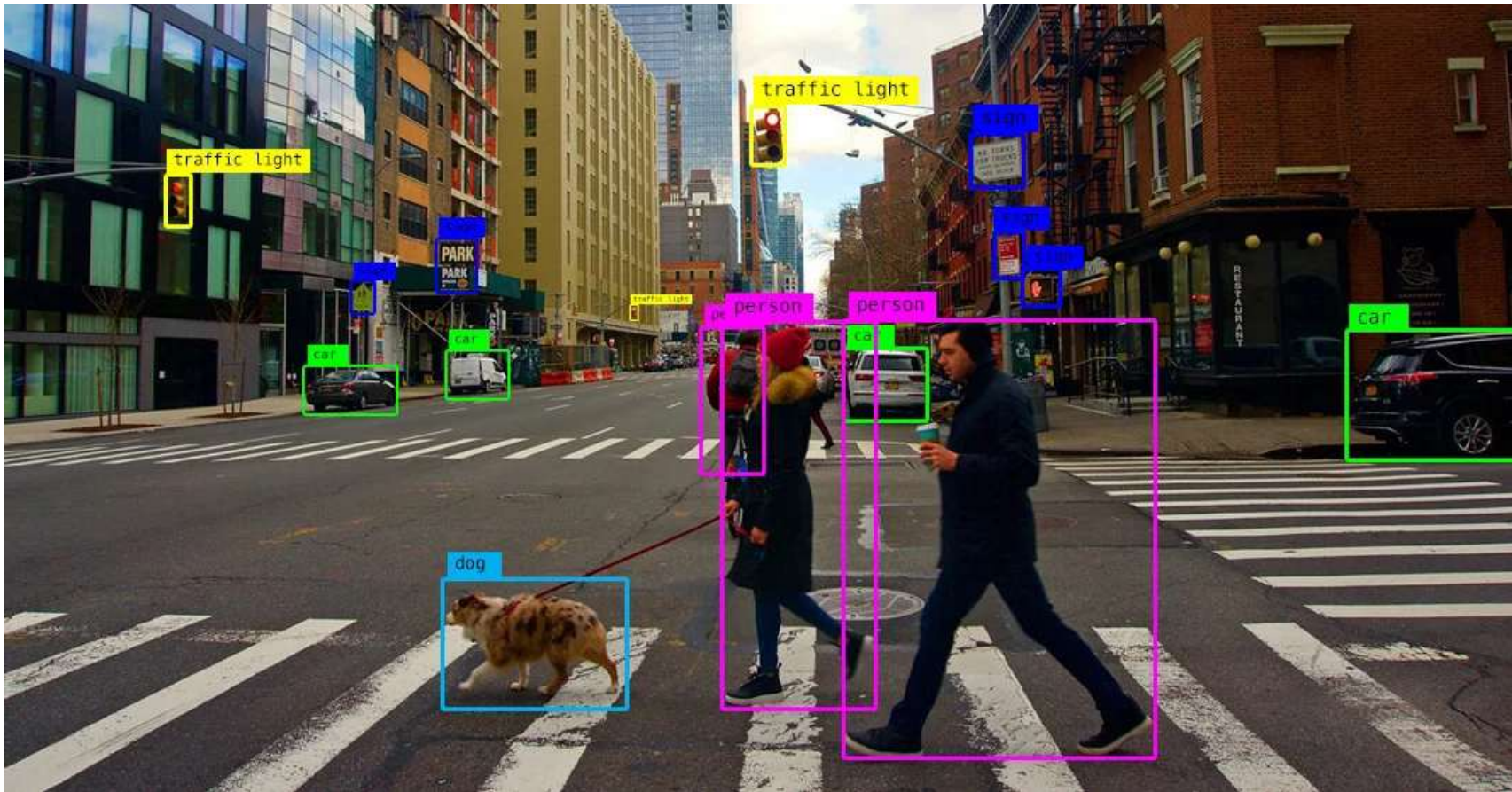
Lecture 1

Introduction to Images





What can the “visual” world tell us?





Can it drive cars?

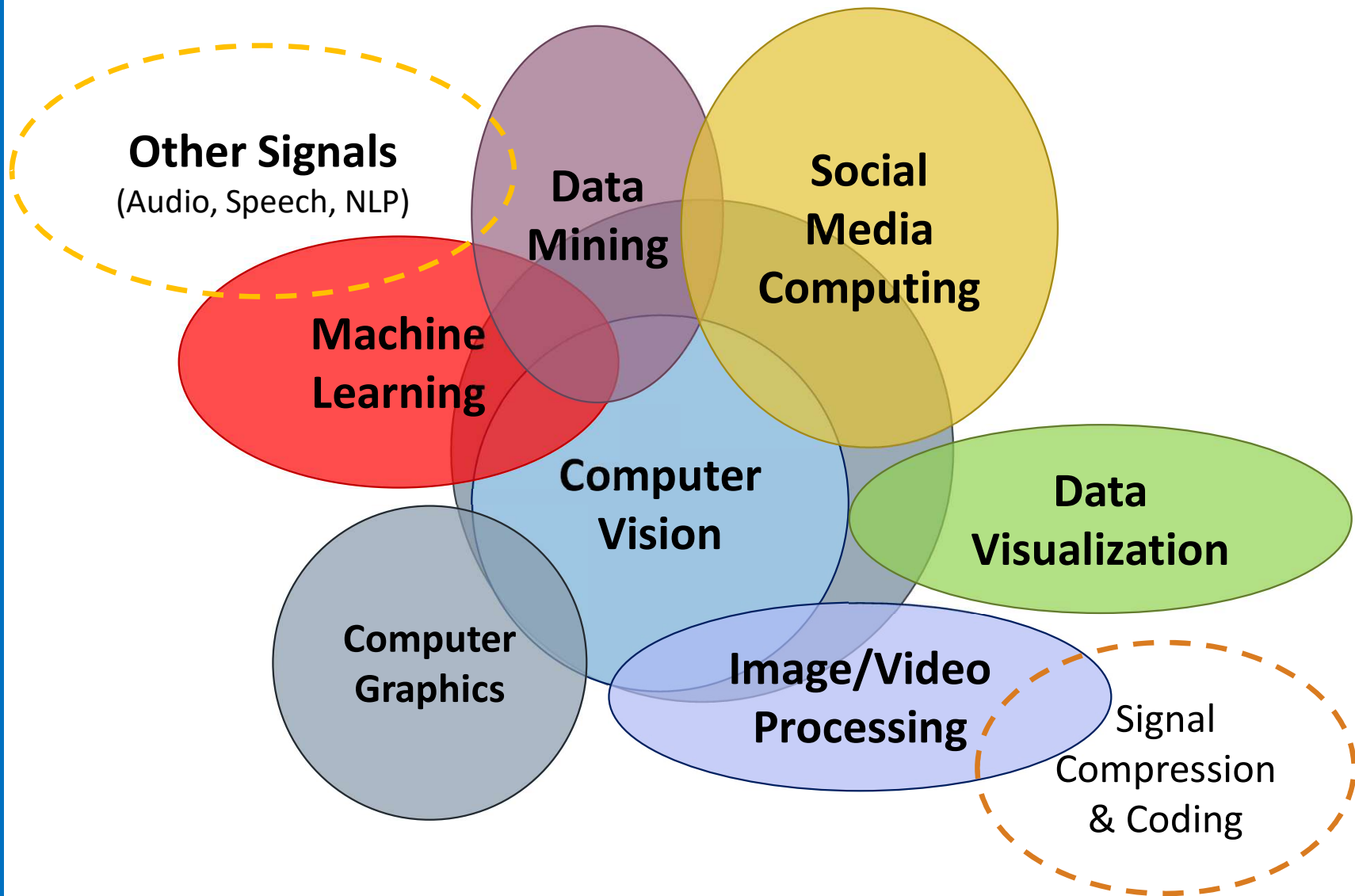




Can it tell us more about others?



Face recognition + Social Network Mining + NLP



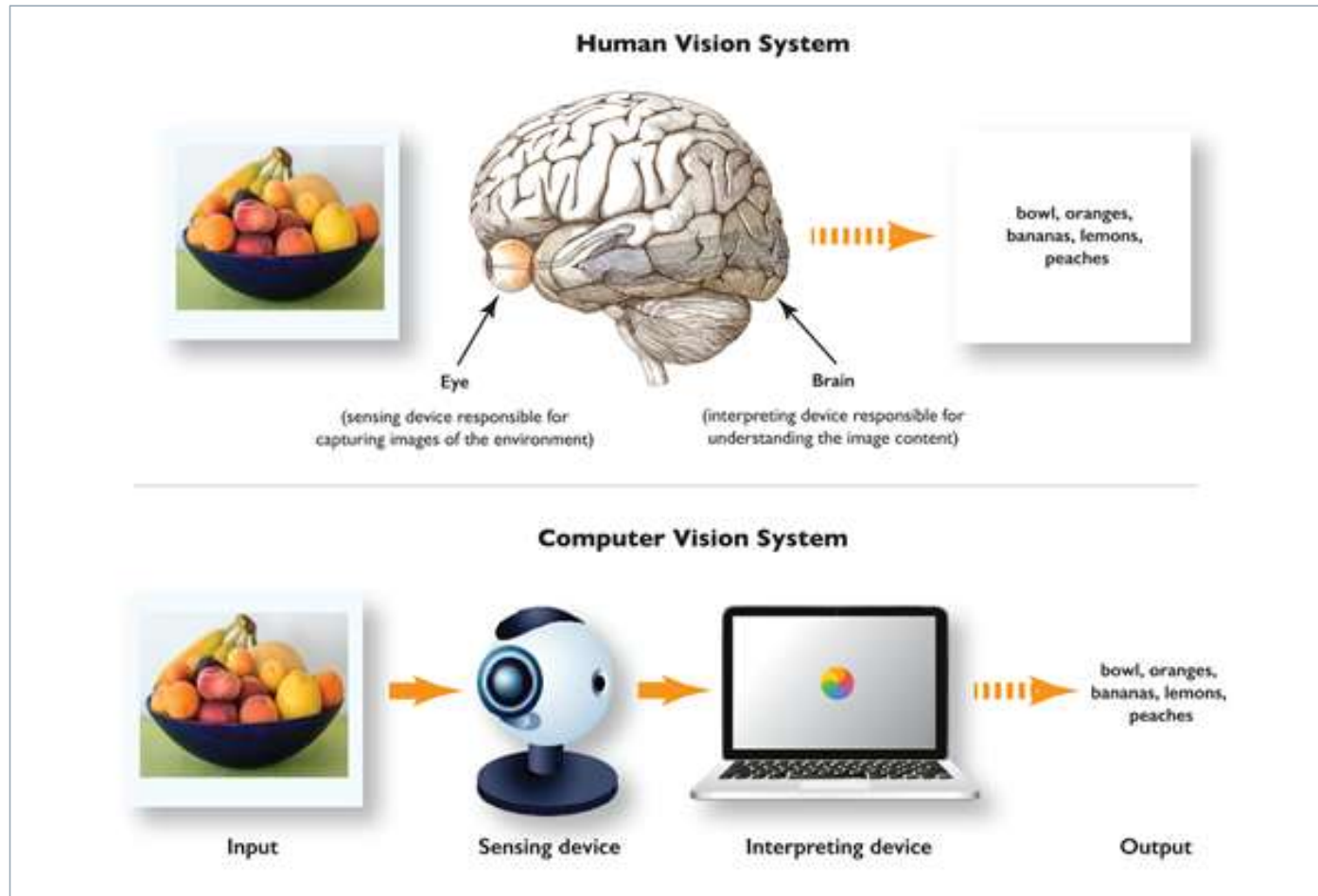


What is computer vision?

- › Computer vision is the branch of Artificial Intelligence responsible for digital systems that detect and process visual information, that is, all kinds of data learned from digital images, videos, and other elements.

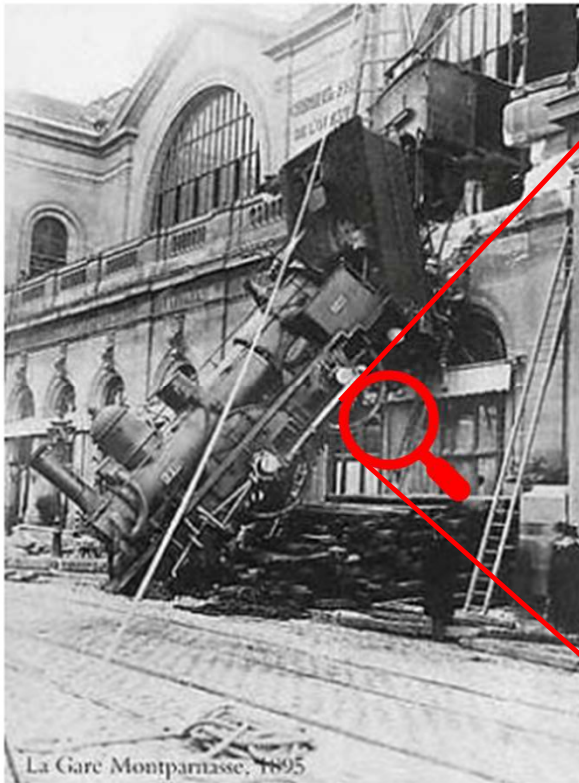


What is computer vision?





Computer Vision: The Goal



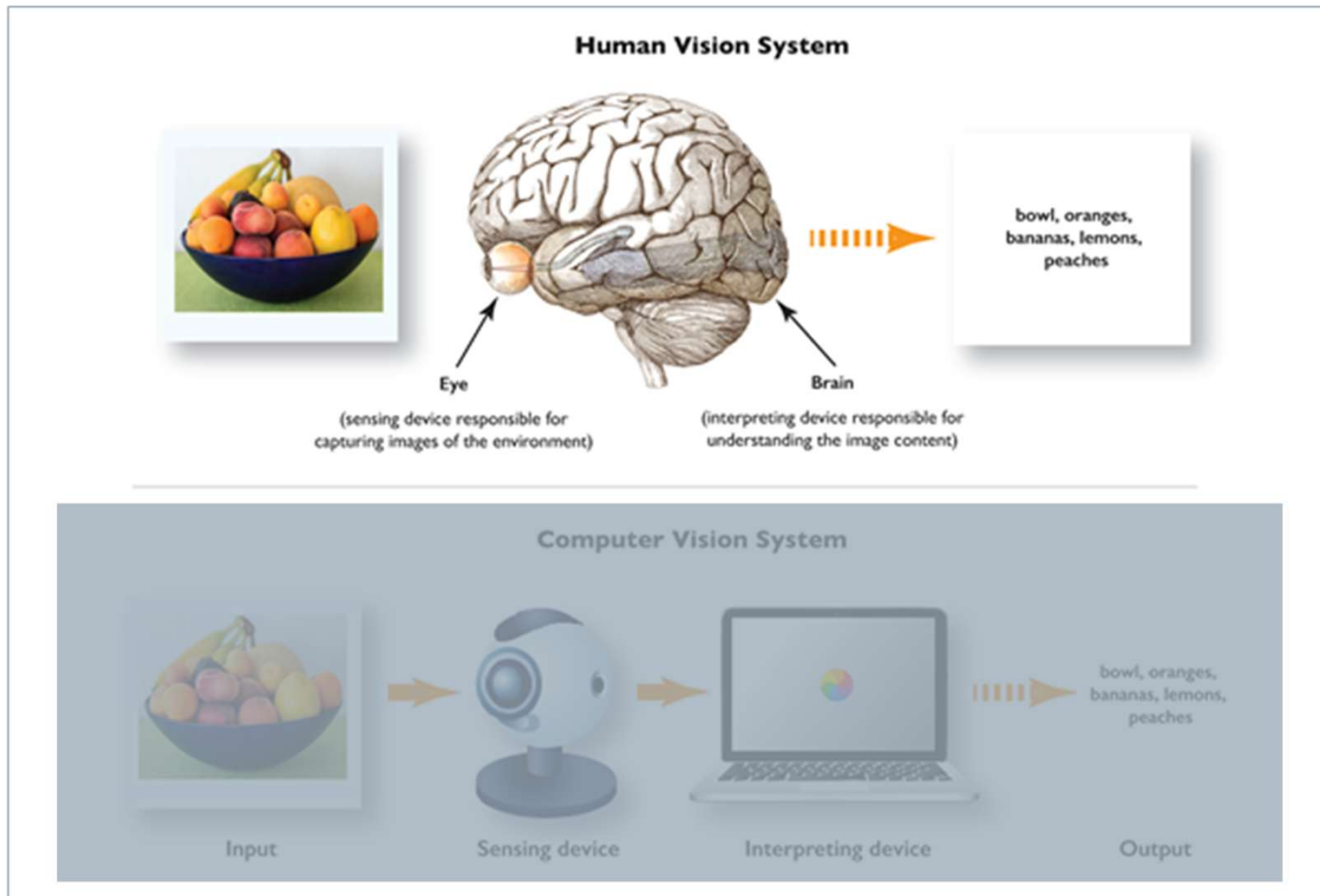
What we see

0	3	2	5	4	7	6	9	8
3	0	1	2	3	4	5	6	7
2	1	0	3	2	5	4	7	6
5	2	3	0	1	2	3	4	5
4	3	2	1	0	3	2	5	4
7	4	5	2	3	0	1	2	3
6	5	4	3	2	1	0	3	2
9	6	7	4	5	2	3	0	1
8	7	6	5	4	3	2	1	0

What the computer sees

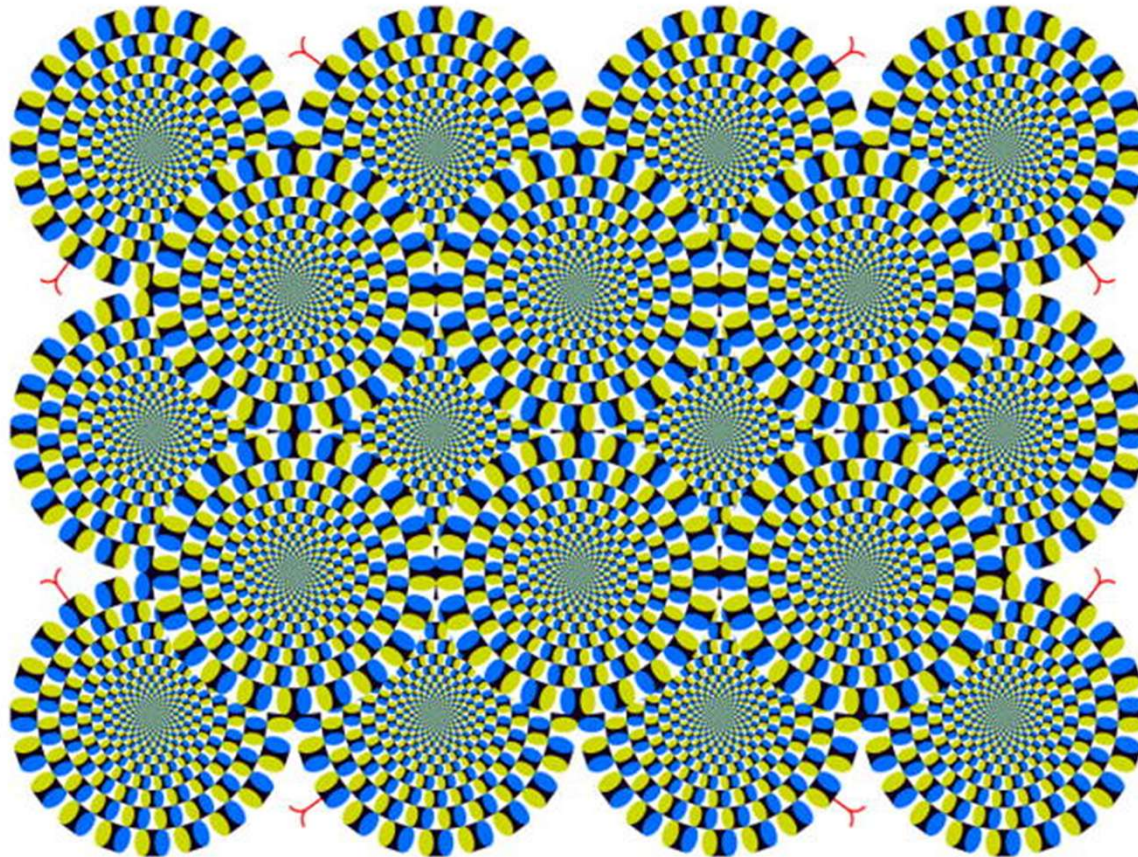


What is the human vision system?



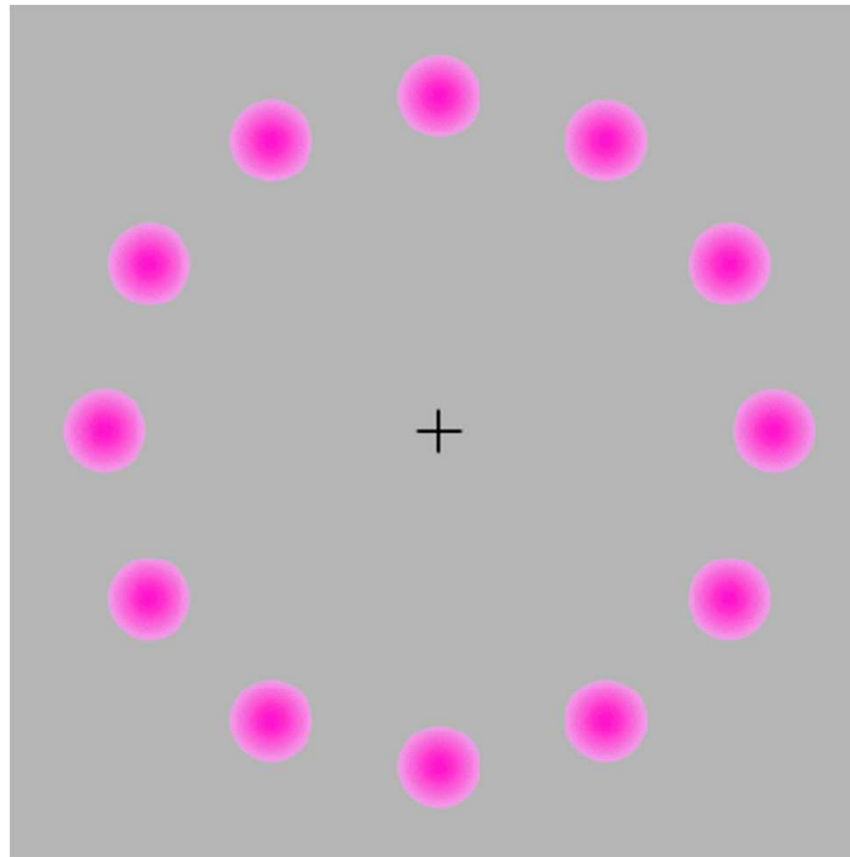


Human vision is superb





Troxler's effect





Is that the same person?

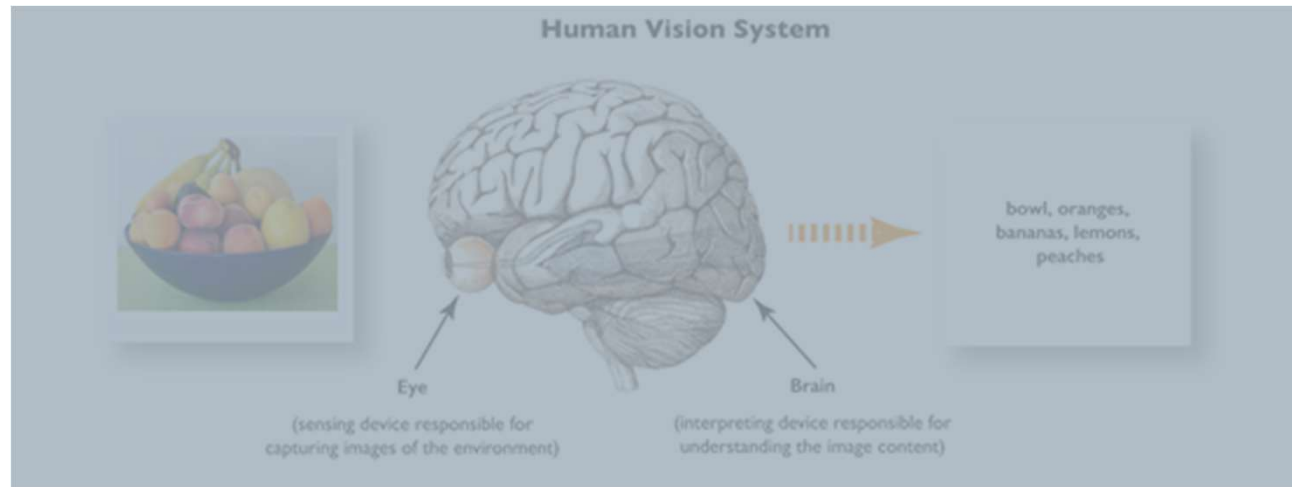


Looks like Ex-President Bill Clinton and his Vice President Al Gore, right?

Wrong... It's Clinton's face twice, with two different haircuts.



What is computer vision?





Fields that deal with Images/Videos

- › **Computer Graphics**: Creation and synthesis of images
- › **Image/Video Processing**: Enhancement or manipulation of an image or video
 - Image $\Rightarrow$ Image | Video $\Rightarrow$ Video
- › **Computer Vision**: Analysis and understanding of image/video content
 - Image/Video $\Rightarrow$ Information (for further use)



Computer Vision

- › Automatic understanding of images and video
 - 1. Computing properties of the 3D world from visual data (measurement)**



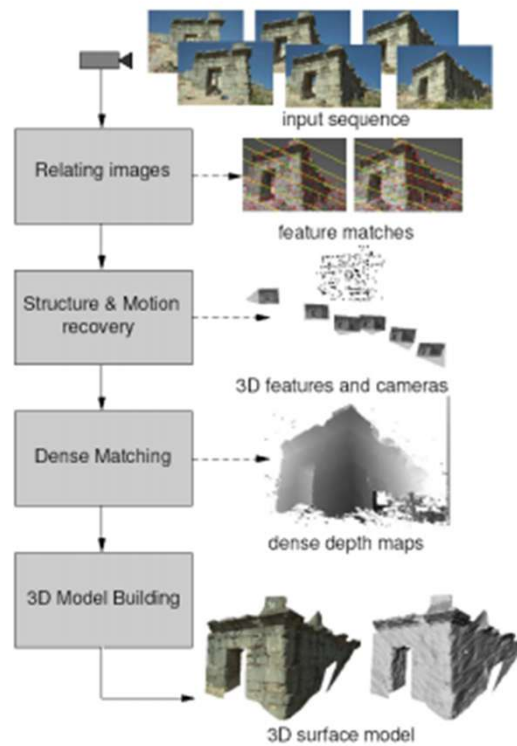
Measurement

Real-time stereo



Wang et al.

Structure from motion



Tracking



Demirdjian et al.

3D reconstruction from image views

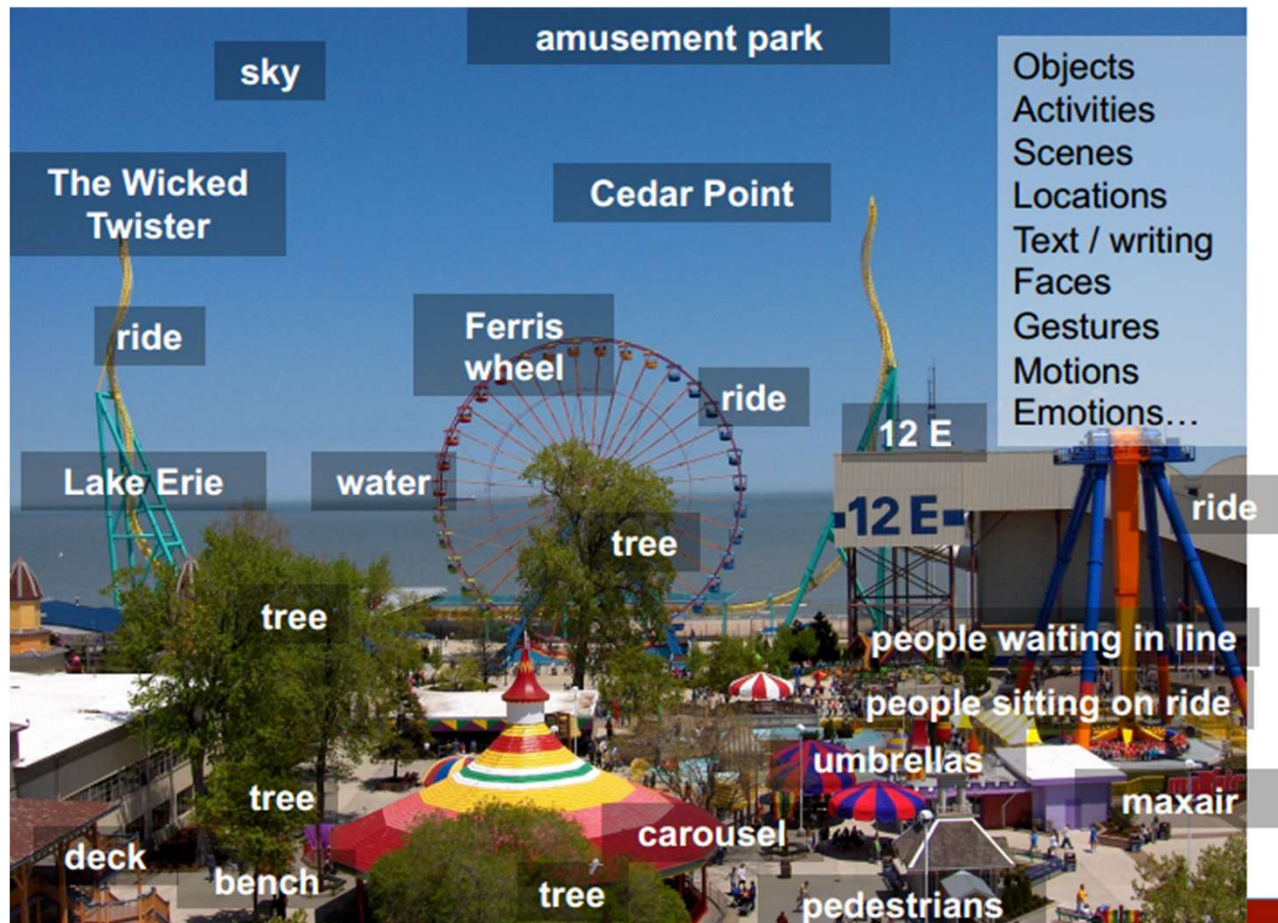


Computer Vision

- › Automatic understanding of images and video
 1. Computing properties of the 3D world from visual data ([measurement](#))
 2. **Algorithms and representations to allow a machine to recognize meaningful entities** (objects, people, scenes, activities, etc.) ([perception and interpretation](#))



Perception and Interpretation



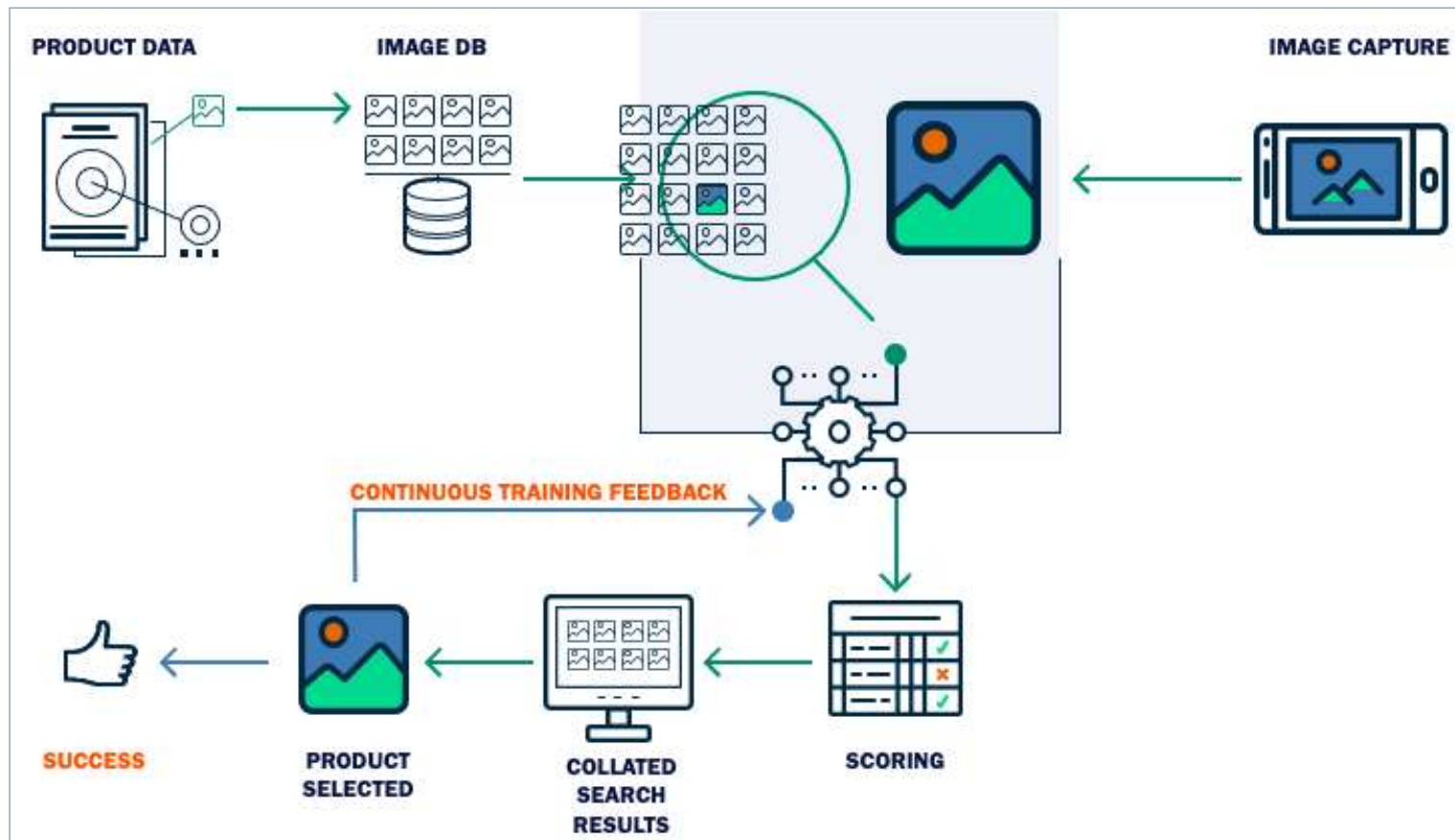


Computer Vision

- › Automatic understanding of images and video
 1. Computing properties of the 3D world from visual data (**measurement**)
 2. Algorithms and representations to allow a machine to recognize meaningful entities (objects, people, scenes, activities, etc.) (**perception and interpretation**)
 3. **Algorithms to mine, search and interact with visual data** (**search and organization**)



Visual Search and Organization





Course Focus

› Automatic understanding of images and video

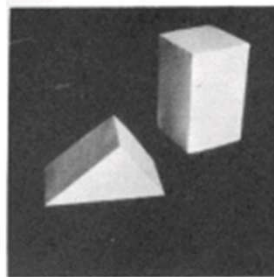
1. Computing properties of the 3D world from visual data (**measurement**)
2. Algorithms and representations to allow a machine to recognize meaningful entities (objects, people, scenes, activities, etc.) (**perception and interpretation**)
3. Algorithms to mine, search and interact with visual data (**search and organization**)

Visual Data and Applications

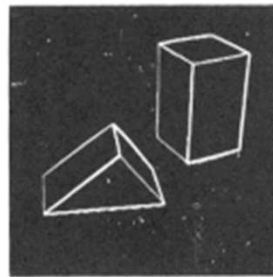




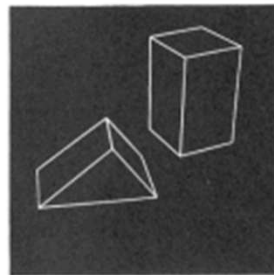
Visual Data in 1963



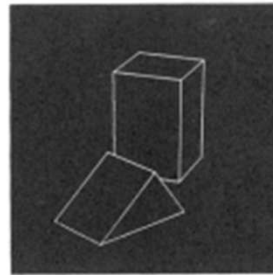
(a) Original picture.



(b) Differentiated picture.

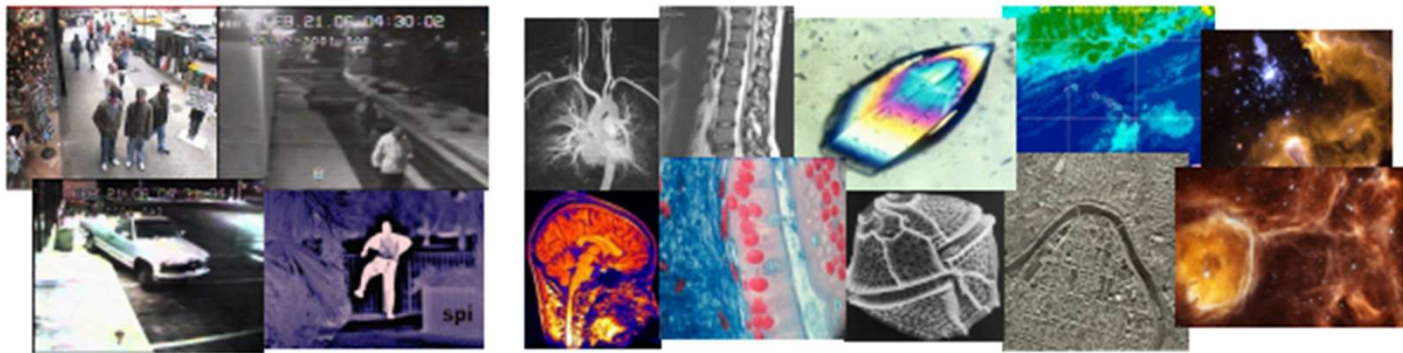


(c) Line drawing.



(d) Rotated view.

L. G. Roberts, [*Machine Perception of Three Dimensional Solids*](#),
Ph.D. thesis, MIT Department of
Electrical Engineering, 1963.



Surveillance and security

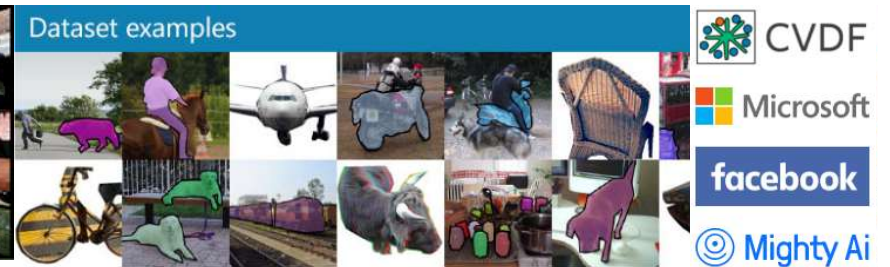
Medical and scientific images



Visual Data in 2021



Labeled Faces in the Wild



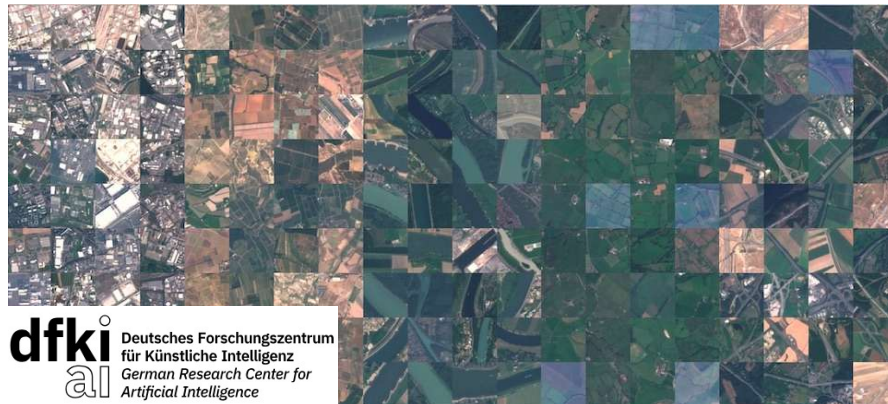
Microsoft Common Objects in Context (COCO)



ImageCLEF/LifeCLEF
Cross Language Evaluation Forum (CLEF)



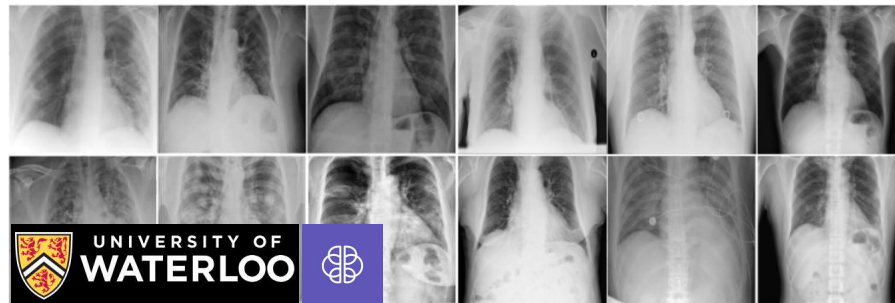
Visual Data in 2024



EuroSAT: Land Use and Land Cover



BAID: Boldbrush Artistic Image Dataset



COVIDx: Chest X-Ray Dataset



Conceptual Captions Dataset



Special effects: Shape & motion capture





Urban modeling: Microsoft Photosynth



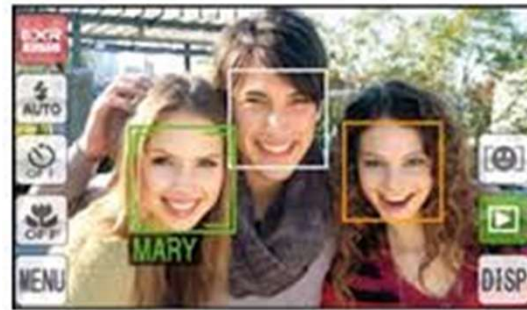
Microsoft Pix

[Pix camera app for iOS.](#)

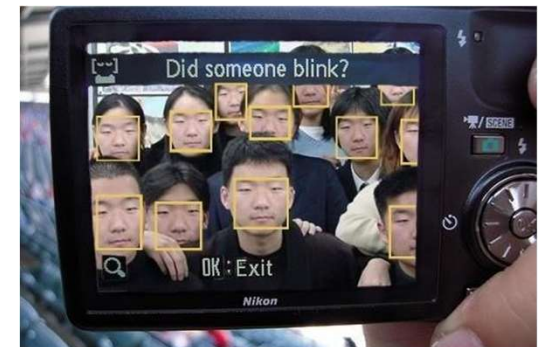


Face detection

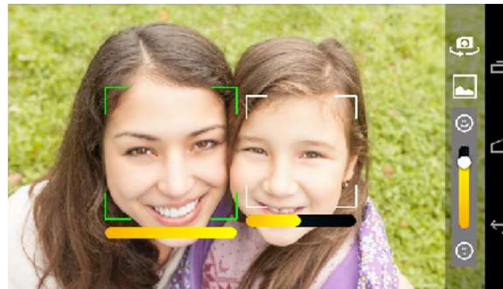
- › Many digital cameras and smartphone cameras can detect faces



Name recognition and
Priority Ranking (Fujifilm)



Blink detector (Nikon)



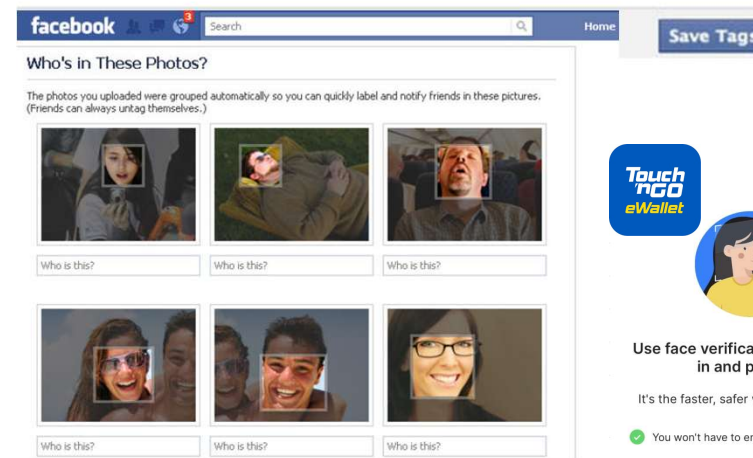
Smile detector (Android camera app, Sony)





Face Recognition

- › FB can “learn” who these people are based on tags
- › Online payment platforms can verify identity
- › Airport biometric scan for luggage drop, boarding, immigration, etc.



Use face verification for future log in and payment?

It's the faster, safer way to log in and pay!

✓ You won't have to enter your PIN

By tapping the "Use Face Verification" button, you acknowledge its usage in accordance with the User [Terms and Conditions](#).

Use Face Verification

Maybe Later



Biometrics

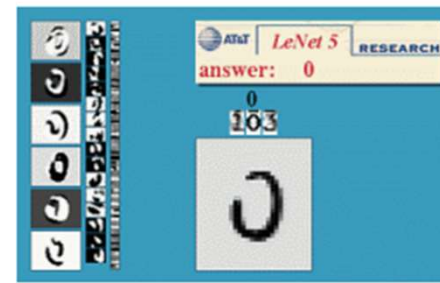
- › Iris scan
- › Fingerprint





Optical Character Recognition (OCR)

- › OCR technology converts scanned docs to text



Digit Recognition, AT&T Labs



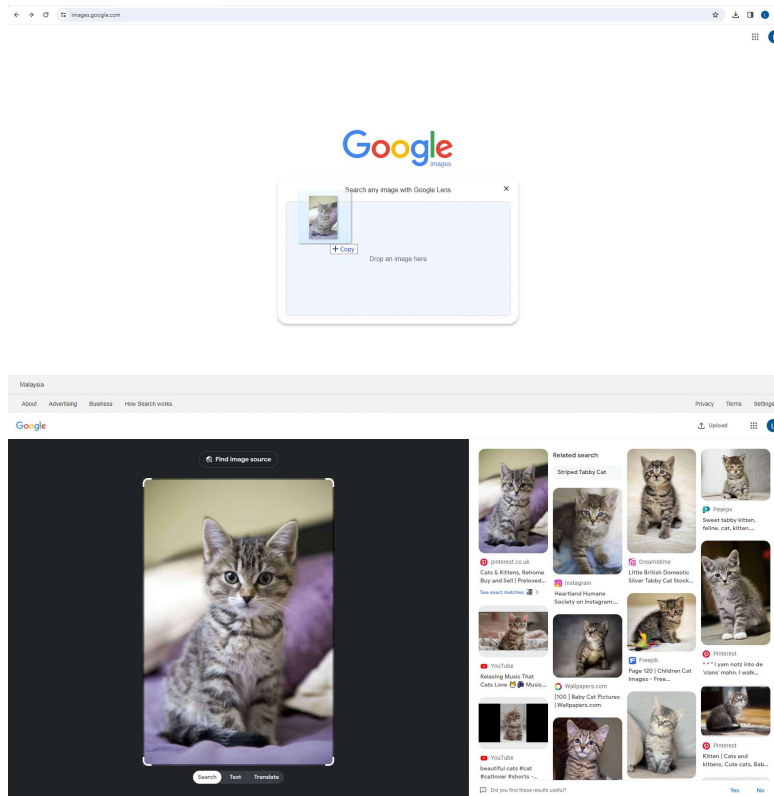
License plate recognition

Production line visual inspection at Vitrox



Image Search / Retrieval

› Google Images



Query
text/image

Red Flower



[27862](#) 0.92 4



[27826](#) 0.92 6



[27887](#) 0.92 8



[27818](#) 0.92 7



[27878](#) 0.91 4



[52385](#) 0.91 5



[3962](#) 0.91 5



[27859](#) 0.91 4



[52307](#) 0.90 10



[27849](#) 0.90 4



[27821](#) 0.90 6



[15723](#) 0.90 6



Automotive Safety

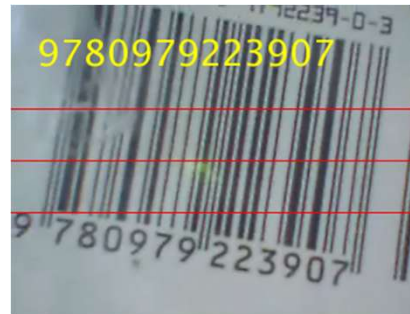
- › Vision systems are found in many car models
 - E.g. Subaru, BMW, Volvo, Lexus, Tesla.





Retail

- › Auto-checkouts
- › Customer tracking



[Amazon Go: The Future Store](#)



<https://www.mashgin.com/>



Vision-based Interaction and Games





Space Exploration



Panoramic view captured by NASA's Mars Exploration Rover Spirit in 2007



Other Applications

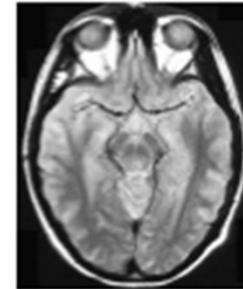
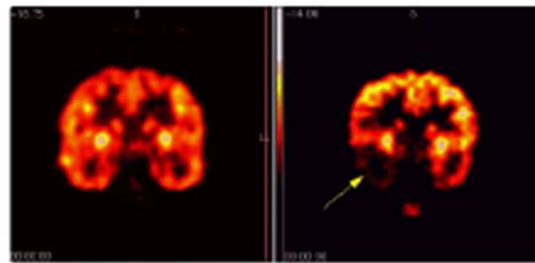
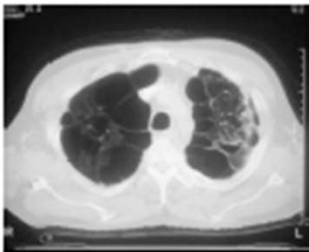
Computer vision
Image processing
(Low-level manipulation of images)





Medical Image Processing

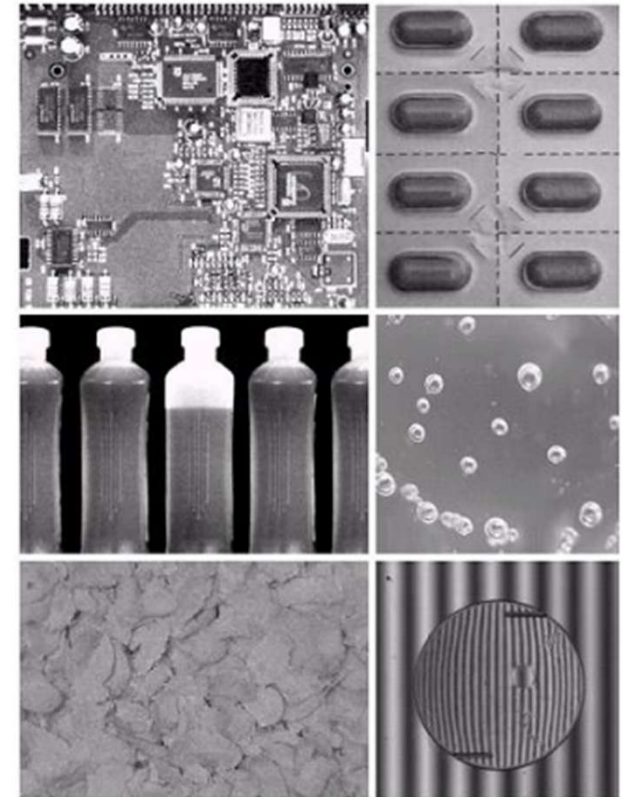
- › Ultra Sound (US)
- › Magnetic Resonance Imaging (MRI)
- › Positron Emission Tomography (PET)
- › Computer Tomography (CT)
- › X-Ray





Automated Industrial Inspection

- › **Circuit board** – missing parts
- › **Pill container** – missing pills
- › **Bottles** – filled up levels
- › **Bubbles in clear-plastic product** – detect unacceptable air pockets
- › **Cereal** – inspection for color, presence of burnt flake
- › **Image of replacement lens for human eye** – inspection of damaged implants





Satellite Image Processing

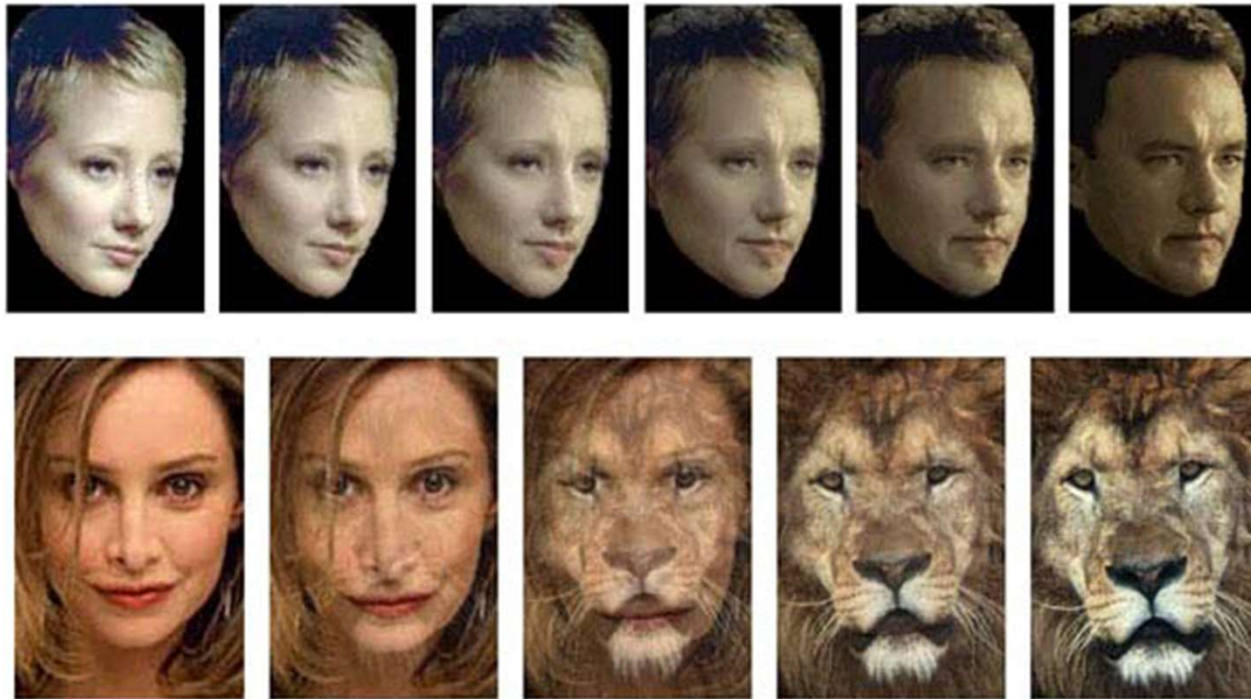
- › Remote sensing
- › Climate study
- › Disaster alert - Flood monitoring, etc.
- › Geology
- › Land resource analysis



New York (from Landsat-5 TM)



Image Morphing

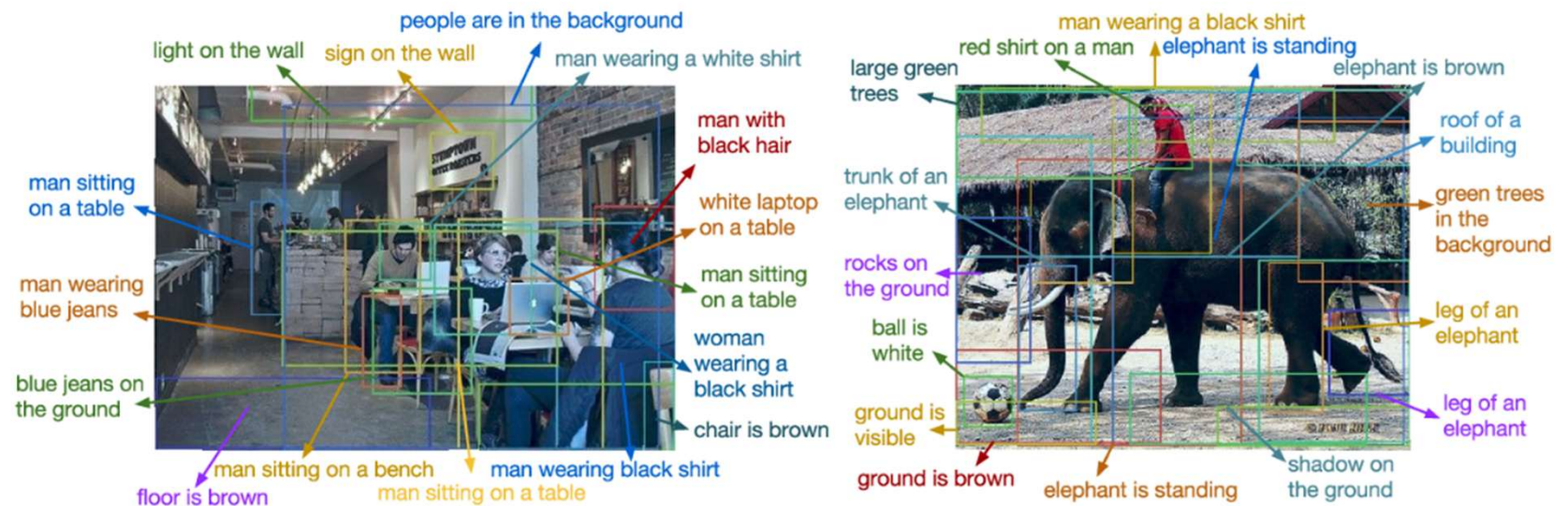


[Morphing Video: From Washington to Trump](#)



New Horizons

› Visual Captioning



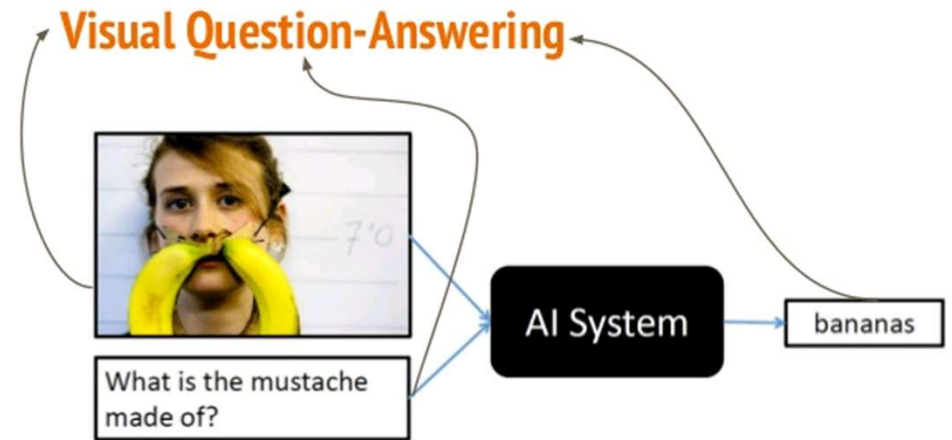
JOHNSON, J., KARPATY, A., & FEI-FEI, L. (2016). DENSECAP: FULLY CONVOLUTIONAL LOCALIZATION NETWORKS FOR DENSE CAPTIONING. IN *PROCEEDINGS OF THE IEEE CONFERENCE ON COMPUTER VISION AND PATTERN RECOGNITION* (PP. 4565-4574).



New Horizons

› Visual Question-Answering

<https://visualqa.org/>



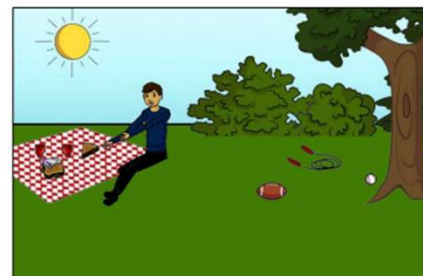
Antol, S., Agrawal, A., Lu, J., Mitchell, M., Batra, D., Lawrence Zitnick, C., & Parikh, D. (2015). Vqa: Visual question answering. In Proceedings of the IEEE International Conference on Computer Vision (pp. 2425-2433).



What color are her eyes?
What is the mustache made of?



How many slices of pizza are there?
Is this a vegetarian pizza?



Is this person expecting company?
What is just under the tree?

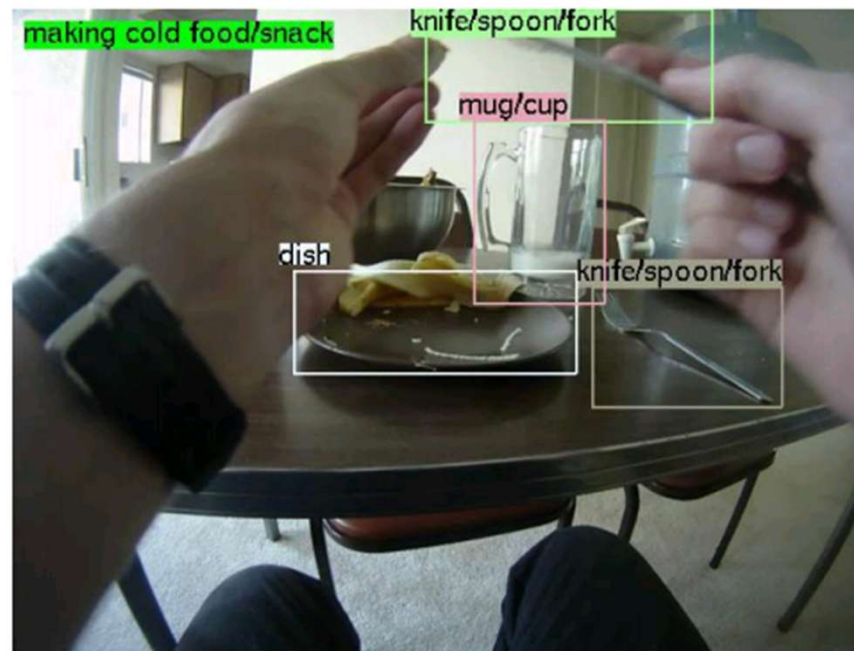


Does it appear to be rainy?
Does this person have 20/20 vision?



New Horizons

› Egocentric (“first person”) vision

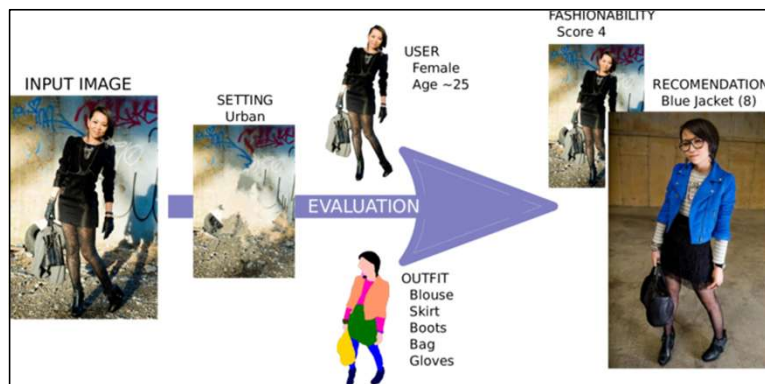


Activities of Daily Living – Hamed Pirsavash et al.



New Horizons

› Fashion Mining and Recommendations



Clothes Segmentation

(<https://esslab.jp/~ess/en/research/fashion/>)

Fashion Style Representation

(<https://esslab.jp/~ess/en/research/stylenet/>)

Virtual Fashion Try-On
(<https://github.com/FashionMirror/FashionMirror>)

Source Human	Try-on Clothes	Target Pose	CPVTON + GFLA	ACGPN + GFLA	FashionOn	VTNCAP	FWGAN	Ours



Why is it challenging to understand visual data?

- › **Ill-posed problem**: Real world much more complex than what we can measure in images
 - $3D \Rightarrow 2D$
- › Can “reconstruct” but **impossible** to “invert” image formation process



Many Problems



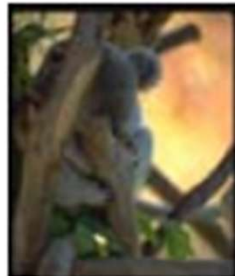
Illumination



Object pose



Clutter



Occlusions



**Intra-class
appearance**



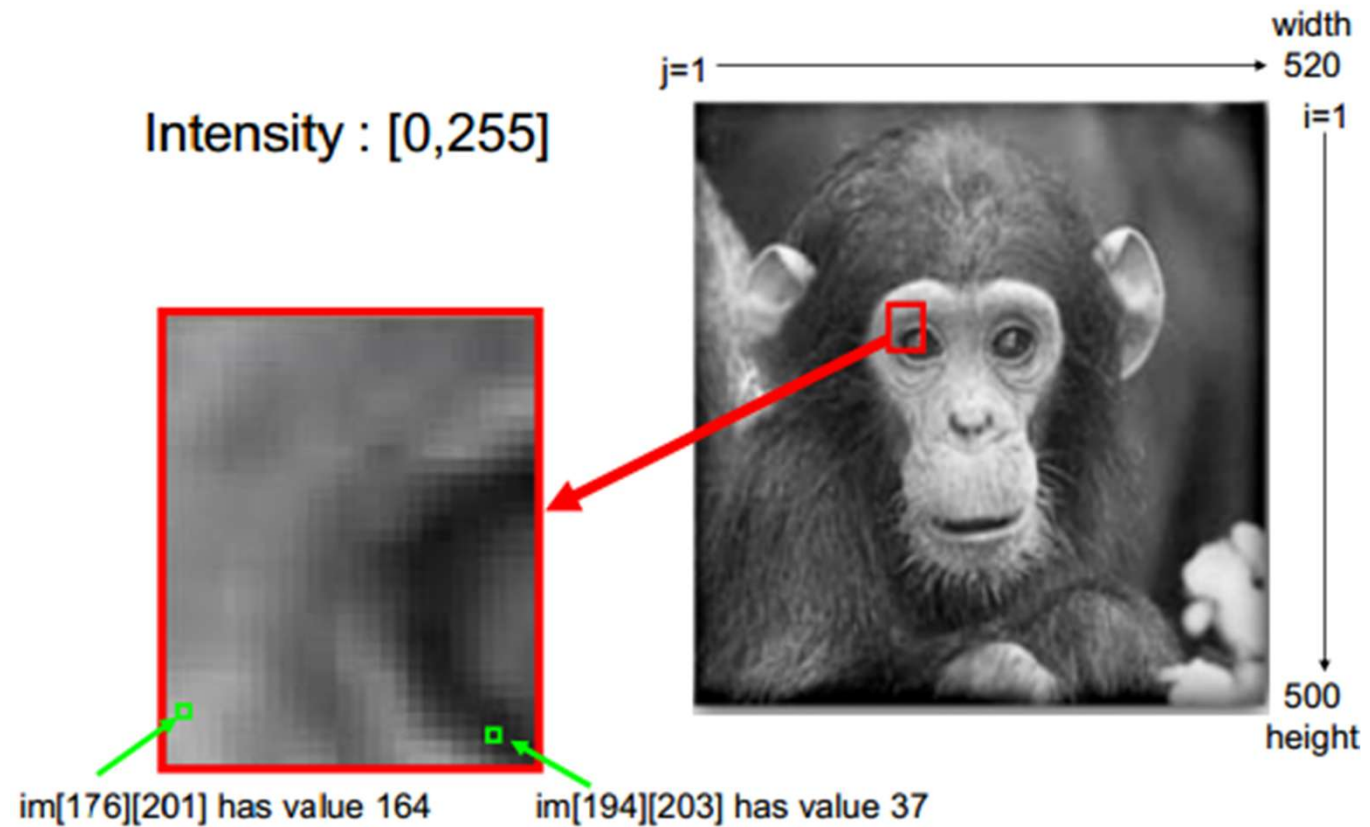
Viewpoint

Topics Preview





Digital Images





Digital Images

Color images,
RGB color
space



R



G



B



Image Processing Tasks

- › Manipulating values
- › Filtering



Observed Image



Enhanced Image
(sharpening)



Image Processing Tasks

- › Restoration
- › Enhancement



Degraded Image



Restored Image



Degraded Image
(salt and pepper noise)

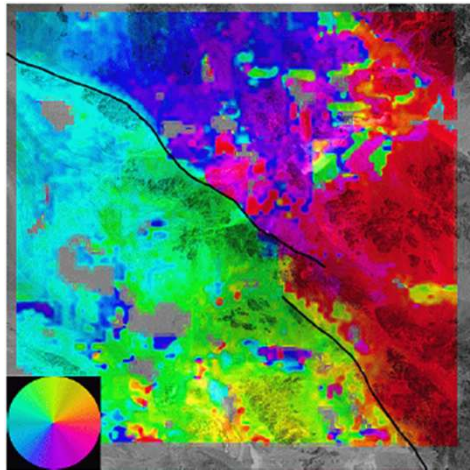


Noise reduced Image
(after median filtering)



Image Processing Tasks

- › Edge Detection
- › Segmentation



Ground replacement due to earthquake in California, 1992





Colour Image Processing

- › Colour spaces
- › Colour Processing





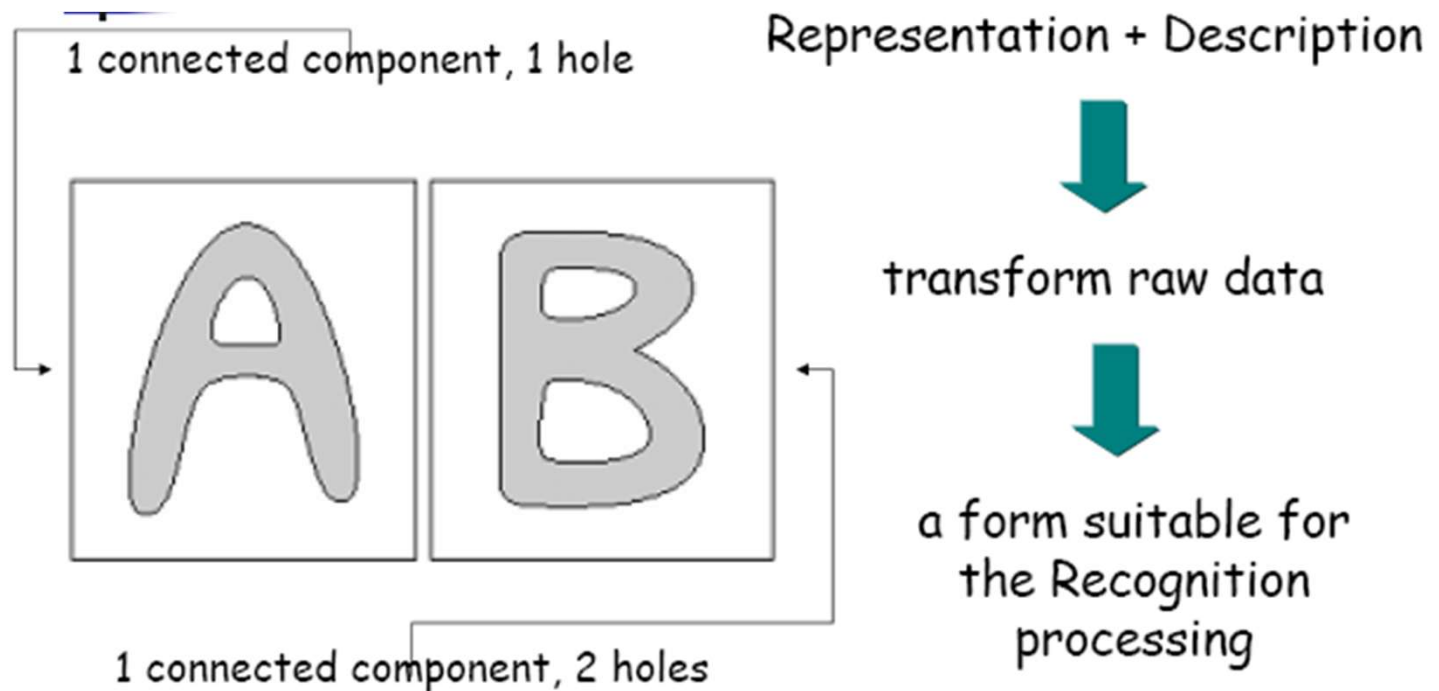
Humans are “still” smarter than computers





Moving up a notch...

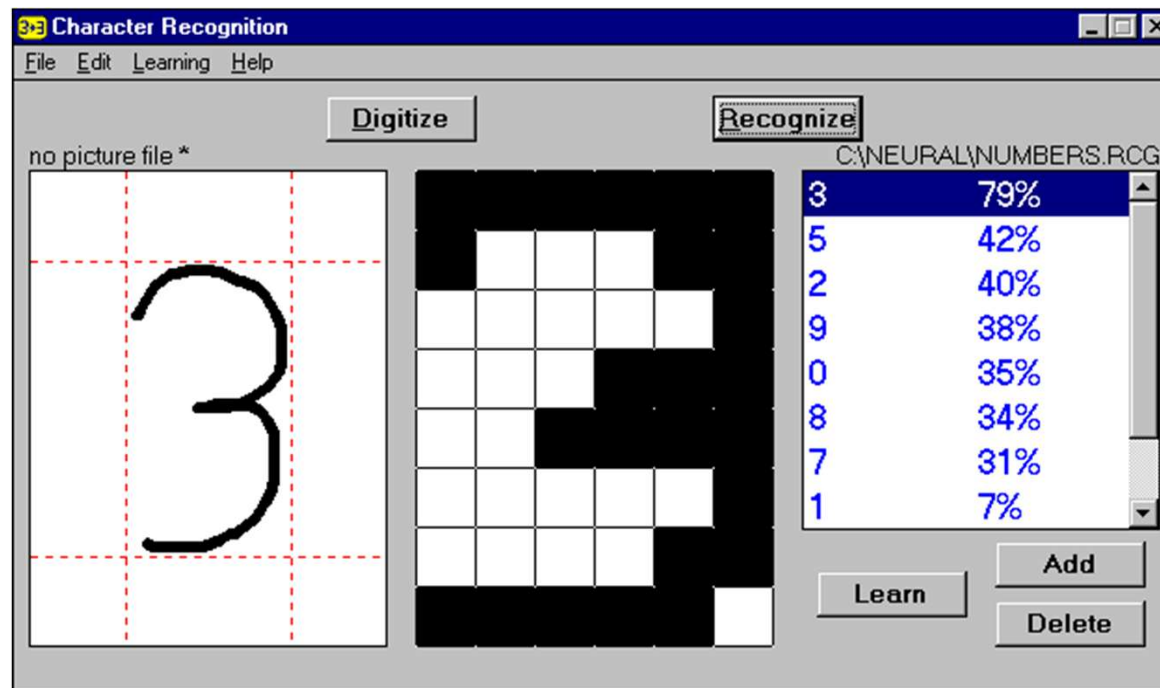
› Representation & Description





...to “higher level” tasks

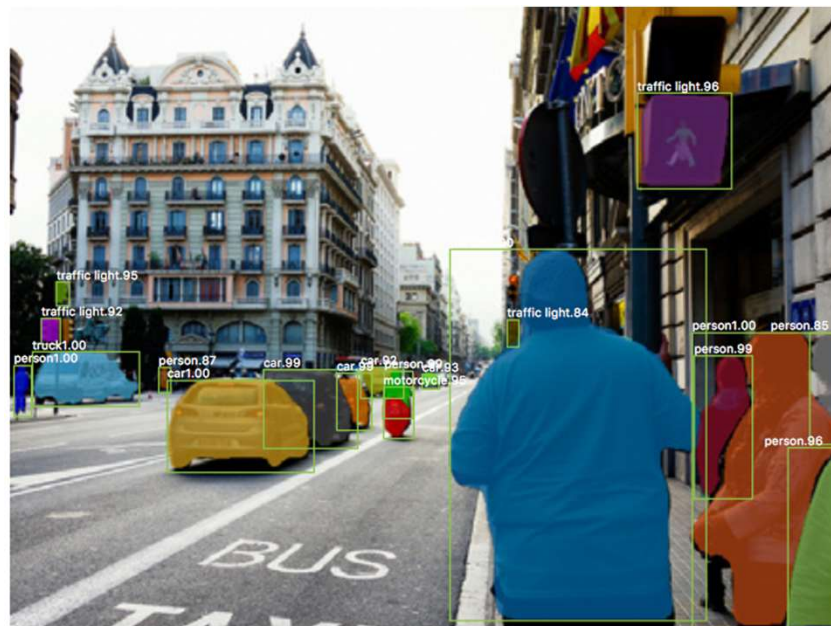
› Recognition & Interpretation





... and even “higher level” tasks

› Recognition & Interpretation



HE, K., GKIOXARI, G., DOLLÁR, P., & GIRSHICK, R. (2017). MASK R-CNN. IN *PROCEEDINGS OF THE IEEE INTERNATIONAL CONFERENCE ON COMPUTER VISION* (PP. 2961-2969).



... and beyond.

› Recognition & Interpretation



ICNet (<https://www.youtube.com/watch?v=qWl9idsCuLQ>)

ZHAO, H., QI, X., SHEN, X., SHI, J., & JIA, J. (2018). ICNET FOR REAL-TIME SEMANTIC SEGMENTATION ON HIGH-RESOLUTION IMAGES. IN *PROCEEDINGS OF THE EUROPEAN CONFERENCE ON COMPUTER VISION (ECCV)* (PP. 405-420).

SUMMARY

- › Image/video processing deals with manipulation
- › Computer vision analyses and deriving understanding
- › Areas include “measurement”, “perception and interpretation”, “search and organization”

